

Environment and Preferences

A two-period OLG economy: Y_t young and O_t old share a fixed housing stock $\bar{H}$.

- Entrants draw income and liquid wealth, (y_i, b_i) , from a fixed distribution F .
- Young choose consumption, housing, fertility, saving, and tenure.
- Old choose consumption, housing, and an estate.

Flow utilities are

$$u_t^Y(c, h, n) = \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta_t \log n,$$

$$u^2(c^2, h^2, e) = \log c^2 + \gamma \log h^2 + \omega_B \log e.$$

Each child needs χ goods and κ housing; e gives warm-glow utility and does not determine entrant wealth.

Old-age utility is discounted by β ; owning adds an independent taste

$\xi_i \sim \text{Logistic}(\bar{\xi}, \sigma_\xi)$.

Housing and Finance

A mortgage finances share ϕ_t of a home bought at price P_t :

$$\underbrace{h \leq h_R^{\max} < h_O^{\max}}_{\text{rent}} \qquad \underbrace{h \leq h_O^{\max}, \quad (1 - \phi_t)P_t h \leq b_i}_{\text{own}}.$$

A buyer may afford the cost of using a home but lack its down payment.

Goods and bonds trade externally at bond price q , with $R_f = q^{-1}$. Rent r_t is paid next period; property tax is $\tau^P P_t$. Rental entry implies

$$qr_t = (1 + q\tau^P)P_t - qP_{t+1} > 0.$$

- Income and rebates arrive after closing; unsecured borrowing is unavailable.
- Old owners may downsize, but cannot buy more or let retained housing.

Household Choices

Taking prices and rebates T_t as given, households choose within each tenure.

Young households

$$W_t^R(i) = \max_{c,h,n,a'} u_t^Y + \beta V_{t+1}^R(R_f a')$$

$$c + a' + q r_t h = y_i + b_i + T_t,$$

$$h \leq h_R^{\max}, \quad a' \geq 0.$$

$$W_t^O(i) = \max_{c,h,n,a'} u_t^Y + \beta V_{t+1}^O(a_{t+1}, h)$$

$$\begin{aligned} c + a' + [(1 - \phi_t) + q \tau^P] P_t h \\ = y_i + b_i + T_t, \end{aligned}$$

$$a_{t+1} = R_f a' - R_f \phi_t P_t h.$$

Owners also satisfy the down payment, $h \leq h_O^{\max}$,

and $a' \geq 0$.

Own if $W_t^O(i) + \xi_i \geq W_t^R(i)$; this determines the owner share $\pi_t^O(i)$.

Old households

$$V_t^R(a) = \max_{c^2, h^2, e} u^2$$

$$c^2 + qe + q r_t h^2 = a + T_t,$$

$$0 < h^2 \leq h_R^{\max}.$$

$$V_t^O(a, H) = \max_{c^2, h^2, e} u^2$$

$$\begin{aligned} c^2 + qe + q r_t h^2 \\ = a + P_t H + T_t, \end{aligned}$$

$$0 < h^2 \leq H, \quad e \geq P_{t+1} h^2.$$

Here a is financial wealth, H inherited housing,

and e the estate in next-period goods.

Competitive Equilibrium

Given policies, F , and (Y_0, O_0, G_0) , an equilibrium is a sequence of prices (P_t, r_t) , rebates T_t , household choices, cohort sizes and old distributions satisfying:

1. Young and old optimize using equilibrium prices; tenure follows the value comparison.
2. Rental entry: $qr_t = (1 + q\tau^P)P_t - qP_{t+1} > 0$.
3. Housing clears and property-tax revenue is rebated:

$$Y_t \bar{h}_t^Y + O_t \bar{h}_t^O = \bar{H}, \quad (Y_t + O_t)T_t = q\tau^P P_t \bar{H}.$$

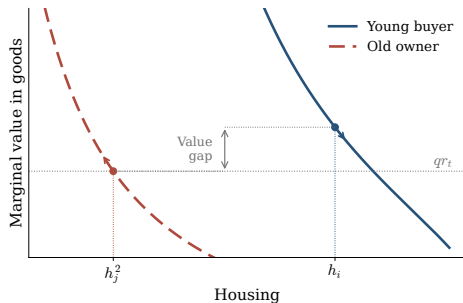
4. Cohorts follow $Y_{t+1} = \nu \bar{n}_t Y_t$ and $O_{t+1} = Y_t$.
5. G_{t+1} follows from young saving, mortgage debt, housing, and tenure.

Bars denote averages across types and tenure; ν converts children to entrants. G_t records old wealth, housing and tenure. Goods and bonds trade externally.

A positive steady state has constant prices, choices and distributions, $Y^* = O^*$, and $\bar{n}^* = 1/\nu$.

Housing Misallocation

For eligible young buyers, only the down-payment constraint binds.



Marginal values in goods:

$$MV_i^Y = \frac{\alpha(c_i - \chi n_i)}{h_i - \kappa n_i},$$

$$MV_j^O = \frac{\gamma c_j^2}{h_j^2}.$$

For an old owner with interior choices:

$$MV_i^Y > qr_t = MV_j^O.$$

Inefficiency. Move a little housing to the young and compensate the old in goods: some young gain and everyone else is equally well off.

Eligible pairs have positive mass; no moving cost. The planner relaxes finance while preserving physical limits, resources, fertility, tenure, estates, future allocations and existing claims.

Housing and Fertility

Relax a young buyer's binding housing limit. Allow fertility to adjust, holding prices and tenure fixed.

Fertility rises if and only if the space benefit exceeds the resource cost:

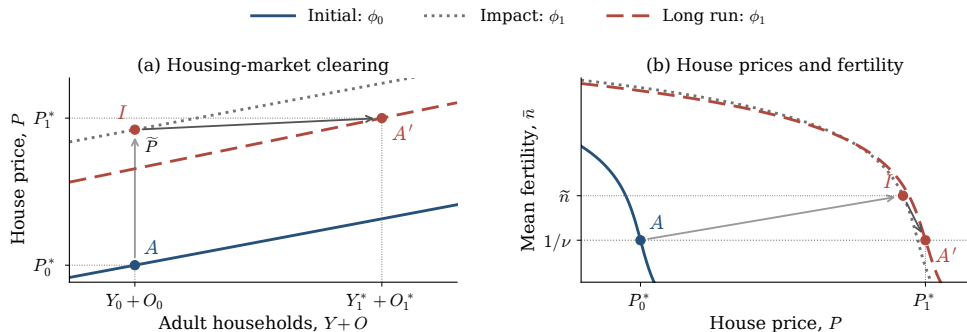
$$\underbrace{\frac{\alpha\kappa}{(h - \kappa n)^2}}_{\text{space benefit}} > \underbrace{\frac{\chi q r_t}{[1 + \beta(1 + \gamma + \omega_B)](c - \chi n)^2}}_{\text{resource cost}}.$$

More space helps accommodate children; paying for it uses other resources.

Positive saving and interior old-age choices; the same constraints bind and no transfers accompany the change.

Equilibrium Transition

Illustration: a small permanent credit relaxation, $\phi_1 > \phi_0$.



Fertility rises at impact and returns to replacement with more households.

Initial/impact: inherited assets and corresponding expectations. Long run: stationary choices. Local schematic curves; adjustment may overshoot.